Chapter 5

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5.1. Let $f : A \to B$ be an integral homomorphism of rings. Show that $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is a closed mapping. (This is a geometric equivalent of (5.10).)

Proof. Take $\mathfrak{b} \trianglelefteq B$, and recall that $f^*(V(\mathfrak{b})) \subset \overline{\mathrm{f^*(V(\mathfrak{b}))}} = V(\mathfrak{b}^c)$ by (Ex. 1.21iii). Now, suppose $\mathfrak{p} \in V(\mathfrak{b}^c)$ is a prime ideal of A, i.e. $\mathfrak{p} \supset \mathfrak{b}^c = f^{-1}(\mathfrak{b})$. Then in particular $\mathfrak{p} \supset \ker f$, so $\mathfrak{p}$ decends to a prime ideal $f(\mathfrak{p})$ of $A/\ker f \cong \operatorname{im} f \subset B$.

$\operatorname{im} f \subset B$ is integral, so (5.10) gives us a prime $\mathfrak{q} \in \operatorname{Spec} B$ with $\mathfrak{q} \cap \operatorname{im} f = f(\mathfrak{p})$. We have that $\mathfrak{q} \supset f(\mathfrak{p}) \supset f(\mathfrak{b}^c)$ implies $\mathfrak{q} \supset \mathfrak{b}^{ce} = \mathfrak{b}$, i.e. $\mathfrak{q} \in V(\mathfrak{b})$. Further,

$$\begin{aligned} f^*(\mathfrak{q}) &= f^{-1}(\mathfrak{q}) \\ &= f^{-1}(\mathfrak{q} \cap \operatorname{im} f) \\ &= f^{-1}(f(\mathfrak{p})) \\ &= \mathfrak{p}, \end{aligned}$$

since $\mathfrak{p} \supset \ker f$. Thus, $\mathfrak{p} \in f^*(V(\mathfrak{b}))$, so in fact $f^*(V(\mathfrak{b})) = \overline{\mathrm{f^*(V(\mathfrak{b}))}}$ is closed. □

5.2. Let B be integral over $A \subset B$, and let $f : A \to \Omega$ be a homomorphism of A into an algebraically closed field Ω. Show that f can be extended to a homomorphism of B into Ω.

Proof. $f : A \to \Omega$ corresponds to a maximal ideal $\mathfrak{p} = \ker f \trianglelefteq A$ and an inclusion $\overline{\mathrm{f}} : A/\mathfrak{p} \hookrightarrow \Omega$. (5.10) supplies us with a prime ideal $\mathfrak{q} \trianglelefteq B$ such that $\mathfrak{q} \cap A = \mathfrak{p}$, which is in fact maximal by (5.8). Then $\pi : B \to B/\mathfrak{q}$ restricts to a map

$$\pi|_A : A \to \frac{A}{\mathfrak{q} \cap A} = \frac{A}{\mathfrak{p}} \subset \frac{B}{\mathfrak{q}}.$$

(5.6) gives us that $K := B/\mathfrak{q}$ is integral over $k := A/\mathfrak{p}$.

Let $(k, \overline{\mathrm{f}}) \leqslant (C, \overline{\mathrm{g}})$ be a maximal element of the poset Σ of maps from subrings of K to Σ, ordered by restriction. In particular, it is a valuation ring of K, so $K \supset C \supset \overline{\mathrm{k}} = K$ by (5.22), whence $K = C$. Then $g := \overline{\mathrm{g}} \circ \pi : B \to \Omega$ has

$$g|_A = \overline{\mathrm{g}}|_k \circ \pi|_A = \overline{\mathrm{f}} \circ \pi|_A = f.$$

□

5.3. Let $f : B \to B'$ be a morphism of A-algebras, and let C be an A-algbera. If f is integral, prove that $f \otimes 1 : B \otimes_A C \to B' \otimes C$ is integral.

Proof. We have that B' is integral over $f(B) \leqslant B'$. That is, $f(B)[b'_1, \ldots, b'_n]$ is finitely generated for all $b'_1, \ldots, b'_n \in B$. Note as well that $(f \otimes 1)(B \otimes C) \cong f(B) \otimes C$. Then, for arbitrary $\sum_{i=1}^n b'_i \otimes c_i \in B' \otimes_A C$,

$$(f \otimes 1)(B \otimes_A C)[\sum b'_i \otimes c_i] \leqslant f(B)[b'_1, \ldots, b'_n] \otimes_A C[c_1, \ldots, c_n].$$

The inclusions

$$f(B) \otimes_A C \hookrightarrow f(B)[b_1', \ldots, b_n'] \otimes_A C \hookrightarrow f(B)[b_1', \ldots, b_n'] \otimes_A C[c_1, \ldots, c_n]$$

are finite by closure of finiteness under base extension (2.17), so the composition is finite by closure of finiteness under composition (2.16). Then our arbitrary element of $B' \otimes_A C$ is integral over $f(B) \otimes_A C$ by (5.1iii), so that $f \otimes 1$ is integral. $\square$

5.4 Let $A \subset B$ be a subring such that B integral over A. Let $\mathfrak{n}$ be a maximal ideal of B and let $\mathfrak{m} = \mathfrak{n} \cap A$ be the corresponding maximal ideal of A. Is $B_\mathfrak{n}$ necessarily integral over $A_\mathfrak{m}$?

Proof. We will show that this need not hold in general. Consider $A := k[x^2 - 1] \subset k[x] =$. Clearly $k[x]$ is integral over A, as it is a finitely generated A module. Consider the maximal ideal $\mathfrak{n} = (x - 1)$. We have that $\mathfrak{n} \supset (x^2 - 1)$ and $\mathfrak{m} := (x^2 - 1)^c$ is maximal in A, so $\mathfrak{n} \cap A \supset \mathfrak{m}$ with $\mathfrak{n}$ a proper ideal implies $\mathfrak{n} \cap A = \mathfrak{m}$.

Now, suppose that $1/(x + 1) \in k[x]_{(x-1)}$ is integral over $A_\mathfrak{m}$. Then we have $n \geqslant 0$ and $a_0, \ldots, a_n \in A_\mathfrak{m}$ such that $(x + 1)^{-n-1} = a_n(x + 1)^{-n} + \cdots + a_0$. Multiplying by $(x + 1)^{n+1}$, we get

$$1 = a_n(x + 1) + \cdots + a_0(x + 1)^{n+1} \in (x + 1)k[x]_{(x+1)},$$

since $(x + 1) \cap k[x^2 - 1] = (x^2 - 1)$ by the same argument as above.

Note that $(x+1)$ is exactly the kernel of the evaluation map $f : k[x] \to k : x \mapsto -1$. Then $k[x]_{(x+1)} = k[x]_S$ where $S = k[x] - \ker f$, so that $f(S) \subset k^\times$ gives us $k_{f(S)} \cong k$. Hence, we receive a map of localizations

$$k[x]_{(x+1)} \xrightarrow{f} k,$$

under which $1 \mapsto 1$ and $a_n(x + 1) + \cdots + a_0(x + 1)^{n+1} \mapsto 0$. $1 \neq 0$ in k, so this contradicts our assumption. $\square$

5.7 Let A be a subring of B such that $B - A$ is closed under multiplication. Show that A is integrally closed in B.

Proof. Suppose that $b \in B - A$ is integral over A, so that $a_0 = a_1 b + \cdots + a_n b^n + b^{n+1}$ for $a_i \in A$ and minimal $n \geqslant 0$. In fact, we may assume $n > 0$, as otherwise $b \in A$. We have

$$a_0 = b(a_1 + \cdots + a_n b^{n-1} + b^n) \in A$$

implies $a_1 + \cdots + a_n b^{n-1} + b^n = a \in A$, as otherwise $a_0 \in B - A$ by closure. Then $a - a_1 = a_2 b + \cdots + a_n b^{n-1} + b^n$ is of strictly lower degree, contradicting minimality of n. $\square$

5.10. A ring homomorphism $f : A \to B$ is said to have the *going-up property* (resp. the *going-down property*) if the conclusion of the going-up theorem (resp. going-down theorem) holds for B and its subring $f(A)$. Let $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ be the mapping associated with f.

(a) Consider the following three statements:
   (i) f^* is a closed mapping.
   (ii) f has the going-up property.
   (iii) For any prime ideal $\mathfrak{q}$ of B, let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \operatorname{Spec}(B/\mathfrak{q}) \to \operatorname{Spec}(A/\mathfrak{p})$ is surjective.

Prove that $(i) \implies (ii) \iff (iii)$.

(b) Consider the following three statements:
  (i') f^* is an open mapping.
  (ii') f has the going-down property.
  (iii') For any prime ideal $\mathfrak{q}$ of B, let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \operatorname{Spec}(B_\mathfrak{q}) \to \operatorname{Spec}(A_\mathfrak{p})$ is surjective.

Prove that (i') $\Longrightarrow$ (ii') $\Longleftrightarrow$ (iii').

Proof. (a) We begin by showing that (ii) $\Longleftrightarrow$ (iii). As in the proof of the going-up theorem, it suffices to consider the case of $\mathfrak{p} \subset \mathfrak{p}' \trianglelefteq A$ and $\mathfrak{q} \trianglelefteq B$ with $\mathfrak{q}^c = \mathfrak{p}$. Applying Spec to the following diagram,

$$\begin{tikzcd} B \arrow[d, two heads] & A \arrow[l, "f"'] \arrow[d, two heads] \\ B/\mathfrak{q} & A/\mathfrak{p} \arrow[l, "\overline{f}"'] \end{tikzcd}$$

we receive

$$\begin{tikzcd} \operatorname{Spec} B \arrow[r, "f^*"] & \operatorname{Spec} A \\ \operatorname{Spec}(B/\mathfrak{q}) \arrow[u, hook] \arrow[r, "f^*"] & \operatorname{Spec}(A/\mathfrak{p}). \arrow[u, hook] \end{tikzcd}$$

Using (1.21iii,iv) and $f^*(\mathfrak{q}) = \mathfrak{p}$, we can restrict to

$$\begin{tikzcd} V(\mathfrak{q}) \arrow[r, "f^*"] & V(\mathfrak{p}) \\ \operatorname{Spec}(B/\mathfrak{q}) \arrow[u, "\wr"] \arrow[r, "f^*"] & \operatorname{Spec}(A/\mathfrak{p}). \arrow[u, "\wr"] \end{tikzcd}$$

Then there exists $\mathfrak{q}' \supset \mathfrak{q}$ with $f^{-1}(\mathfrak{q}') = \mathfrak{p}'$ if and only if $\overline{\mathfrak{p}}' \in \operatorname{im} f^*$. Hence, surjectivity of $f^* : \operatorname{Spec}(B/\mathfrak{q}) \to \operatorname{Spec}(A/\mathfrak{p})$ holds for all prime ideals $\mathfrak{q} \trianglelefteq B$ if and only if $f : A \to B$ possesses the going-up property.

Now, we will show that (i) $\Longrightarrow$ (iii). Suppose f^* is a closed mapping, and recall the last diagram from above. By (1.21iii) once more, we have that

$$\begin{aligned} f^*(V(\mathfrak{q})) &= \overline{\mathrm{f^*(V(\mathfrak{q}))}} \\ &= V(\mathfrak{q}^c) \\ &= V(\mathfrak{p}), \end{aligned}$$

giving surjectivity of $f^* : \operatorname{Spec}(B/\mathfrak{q}) \to \operatorname{Spec}(A/\mathfrak{p})$.

(b) We begin by showing that (ii') $\Longleftrightarrow$ (iii'). Once again, we need only consider the case of prime ideals $\mathfrak{p}' \subset \mathfrak{p} \trianglelefteq A$ with $\mathfrak{q} \trianglelefteq B$ such that $\mathfrak{q}^c = \mathfrak{p}$. Fixing $\mathfrak{q}$ and $\mathfrak{p} = \mathfrak{q}^c$, (3.11iv) gives us the diagram

$$\begin{tikzcd} \{\mathfrak{q}' \subset \mathfrak{q}\} \arrow[r, "f^*"] & \{\mathfrak{p}' \subset \mathfrak{q}\} \\ \operatorname{Spec} B_\mathfrak{q} \arrow[u, "\wr"] \arrow[r, "f^*"] & \operatorname{Spec} A_\mathfrak{p}, \arrow[u, "\wr"] \end{tikzcd}$$

whence we conclude our result similarly.

Finally, we show that (i') $\implies$ (ii'). Suppose $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is openFix $\mathfrak{p}' \subset \mathfrak{p} \trianglelefteqslant A$, and $\mathfrak{q} \trianglelefteqslant B$ lying over $\mathfrak{p}$, i.e. with $\mathfrak{q}^c = \mathfrak{p}$. For any $b \in B \setminus \mathfrak{q}$, $f^*(X_b) \ni \mathfrak{p}$ is open, so $\mathfrak{p}' \rightsquigarrow \mathfrak{p}$ implies $\mathfrak{p}' \in f^*(X_b)$. This supplies us with $\mathfrak{q}'_b \in X_b$ such that $f^*(\mathfrak{q}'_b) = \mathfrak{p}'$, whence we receive

$$\mathfrak{q}' := \bigcap_{b \in B \setminus \mathfrak{q}} \mathfrak{q}'_b \subset \mathfrak{q}$$

such that

$$f^*(\mathfrak{q}') = \bigcap_{b \in B \setminus \mathfrak{q}} f^*(\mathfrak{q}'_b) = \mathfrak{p}',$$

as desired. □

5.11 Let $f : A \to B$ be a flat homomorphism of rings. Then f has the going down property.

Proof. This is a corollary of (Ex. 3.18). □

5.16. Let k be a field and let $A \neq 0$ be a finitely generated k-algebra. Then there exist elements $y_1, \ldots y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \ldots, y_r]$.

We shall assume that k is *infinite*. (A different proof is required for the finite case.) From the proof, it follows that $y_1, \ldots, y_r$ may be chosen to be linear combinations of $x_1, \ldots, x_n$. This has the following geometrical interpretation: if k is algebraically closed and X is an affine algbraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L. [Use Exercise (2).]

5.27 Let A,B be two local rings. B is said to *dominate* A if A is a subring of B and the maximal ideal $\mathfrak{m}$ of A is contained in the maximal ideal $\mathfrak{n}$ of B (or, equivalently, if $m = \mathfrak{n} \cap A$).

Let K be a field and let Σ be the set of all local subrings of K. If Σ is ordered by the relation of domination, show that Σ has maximal elements, and that $A \in \Sigma$ is maximal if and only if A is a valuation ring of K.

Proof. Consider the poset

$$\Sigma' = \{(A, f) \mid A \leqslant K,\ f : A \to \overline{\mathrm{K}}\}$$

ordered by restriction. Clearly $\Sigma \subset \Sigma'$ as sets by identifying $A \leqslant K$ with $(A,\ A \hookrightarrow K \hookrightarrow \overline{\mathrm{K}})$. Suppose $(A,\ i) \leqslant_{\Sigma'} (B,\ j)$. Then A is a subring of A', and

$$\mathfrak{m} = \ker i = \ker j|_A = \ker j \cap A = \mathfrak{n} \cap A$$

by (5.19), so that $A \leqslant_\Sigma B$.

Suppose that $A \in \Sigma$ is maximal. If $(A, i) \leqslant_{\Sigma'} (B, g)$, then □

5.28 Let A be an integral domain, K its field of fractions. Show that the following are equivalent:

(a) A is a valuation ring of K;

(b) If $\mathfrak{a}$, $\mathfrak{b}$ are any two ideals of A, then either $\mathfrak{a} \subset \mathfrak{b}$ or $\mathfrak{b} \subset \mathfrak{a}$.

Deduce that if A is a valuation ring and $\mathfrak{p}$ is a prime ideal of A, then $A_\mathfrak{p}$ and $A/\mathfrak{p}$ are valuation rings of their fields of fractions.

Proof. ((a) $\implies$ (b)) Let $\mathfrak{a}, \mathfrak{b} \trianglelefteq A$, and suppose without loss of generality that $\mathfrak{a} \not\subset \mathfrak{b}$. Then fix $f \in \mathfrak{a} \setminus \mathfrak{b}$ and let $0 \neq g \in \mathfrak{b}$. $f/g \notin A$, so A a valuation ring gives $g/f \in A$, whence $g \in \mathfrak{a}$, so that $\mathfrak{b} \subset \mathfrak{a}$.

((b) $\implies$ (a)) Let $x/y \in K$, and consider $(x), (y) \trianglelefteq A$. If $(x) \subset (y)$, then $x \in (y)$ implies $x/y \in A$. If $(y) \subset (x)$, then $y \in (x)$ implies $y/x \in A$.

Now, suppose A is a valuation ring and $\mathfrak{p} \trianglelefteq A$ is a prime ideal. Any $\mathfrak{a}, \mathfrak{b} \trianglelefteq A_\mathfrak{p}$ are extended, and $\mathfrak{a}^c \subset \mathfrak{b}^c$ or $\mathfrak{a}^c \supset \mathfrak{b}^c$ by the above, whence $\mathfrak{a} \subset \mathfrak{b}$ or $\mathfrak{a} \supset \mathfrak{b}$. The ideal lattice of $A/\mathfrak{p}$ is isomorphic to the poset $\{\mathfrak{a} \supset \mathfrak{p}\}$, so it is clear that the second criterion holds. $\square$

5.29 Let A be a valuation ring of a field K. Show that every subring of K which contains A is a localization of A.

Proof. Let B be a subring of K containing A. By (5.18i,ii), B is a valuation ring, and hence local with maximal ideal $\mathfrak{m} \trianglelefteq B$. Let $\mathfrak{n} := A \cap \mathfrak{m}$ be the contraction of $\mathfrak{m}$ by $A \hookrightarrow B$, and consider the localization $A_\mathfrak{n} \subset B$. If $f \in B \setminus A$, then $f^{-1} \in A$, so that $f \in B^\times = B \setminus \mathfrak{m}$ implies $f = (f^{-1})^{-1} \in A_\mathfrak{n}$, whence $B \subset A_\mathfrak{n}$. $\square$

5.30 Let A be a valuation ring of a field K. The group $A^\times$ of units of A is a subgroup of the multiplicative group $K^\times$ of K.

Let $\Gamma = K^\times/A^\times$. If $\xi, \eta \in \Gamma$ are represented by $x, y \in K$, define $\xi \geqslant \eta$ to mean $xy^{-1} \in A$. Show that this defines a total ordering on Γ which is compatible with the group structure (i.e., $\xi \geqslant \eta \implies \xi\omega \geqslant \eta\omega$ for all $\omega \in \Gamma$). In other words, Γ is a totally ordered abelian group. It is called the *value group of* A.

Let $v : K^\times \to \Gamma$ be the canonical homomorphism. Show that $v(x + y) \geqslant \min(v(x), v(y))$ for all $x, y \in K^\times$.

Proof. Clearly $\geqslant$ is reflexive and transitive. If $\xi \geqslant \eta$ and $\eta \geqslant \xi$, then $xy^{-1} \in A$ and $yx^{-1} \in A$ implies $xy^{-1} \in A^\times$, so that $\xi = \eta$ gives us anti-symmetry. It is total by definition of a valuation ring. Now, we show it is compatible with the group structure. Let $\xi \geqslant \eta$, $\omega \in \Gamma$, and take representatives $x, y, z \in K$. Then $(xz)(yz)^{-1} = xy^{-1} \in A$ implies $\xi\omega \geqslant \eta\omega$.

Suppose, without loss of generality, that $v(x) \geqslant v(y)$, i.e. $xy^{-1} \in A$. Then

$$(x + y)y^{-1} = xy^{-1} + 1 \in A$$

implies $v(x + y) \geqslant v(y)$, as desired. Then $d(x, y) := -v(x - y)$ defines an ultrametric on K, thus giving K the structure of a non-archimedean field. $\square$

5.31 Conversely, let Γ be a totally ordered abelian group (written *additively*), and let K be a field. A *valuation of* K *with values in* Γ is a mapping $v : K^\times \to \Gamma$ such that

(a) $v(xy) = v(x) + v(y)$,

(b) $v(x + y) \geqslant \min(v(x), v(y))$,

for all $x, y \in K^\times$. Show that the set of elements $x \in K$ such that $x = 0$ or $v(x) \geqslant 0$ is a valuation ring of K. This ring is called the valuation ring of v, and the subgroup $v(K^\times)$ is the *value group* of v.

Thus, the concepts of valuation ring and valuation are essentially equivalent.

Proof. Define $A := \{x \in K : x = 0 \text{ or } v(x) \geqslant 0\}$. Then $0 \in A$, A is additive by (b), and multiplicative by (a). If $x \in K^\times$, then

$$v(x) + v(x^{-1}) = v(1) = 0$$

implies either $v(x) \geqslant 0$ or $v(x^{-1}) \geqslant 0$. That is, $x \in A$ or $x^{-1} \in A$. We conclude that A is indeed a valuation ring of K. □

5.32 Let Γ be a totally ordered abelian group. A subgroup $\Delta \leqslant \Gamma$ is *isolated* in Γ if, whenever $0 \leqslant \beta \leqslant \alpha$ and $\alpha \in \Delta$, we have $\beta \in \Delta$. Let A be a valuation ring of a field K, with value group Γ. If $\mathfrak{p}$ is a prime ideal of A, show that $v(A \setminus \mathfrak{p})$ is the set of elements $\geqslant 0$ in an isolated subgroup $\Delta \leqslant \Gamma$, and that the mapping so defined of $\operatorname{Spec} A$ into isolated subgroups of Γ is surjective.

If $p \leqslant A$ is prime, what are the value groups of the valuation rings $A/\mathfrak{p}$, $A_\mathfrak{p}$?

Proof. Suppose $\alpha = v(a) \in v(A \setminus \mathfrak{p})$, and let $0 \leqslant \beta \leqslant \alpha$ for $\beta = v(b) \in v(A)$. Then $ab^{-1} \in A$, so $b \in A \setminus \mathfrak{p}$, as otherwise $a = (ab^{-1})b \in \mathfrak{p}$. Hence, $\beta \in v(A \setminus \mathfrak{p})$.

Now, let $\Delta \leqslant \Gamma$ be an isolated subgroup, and consider $\mathfrak{p} := v^{-1}(\Gamma \setminus \Delta) \cup \{0\}$. Additivity is clear. If $b \in v^{-1}(\Gamma \setminus \Delta)$ and $a \in A \setminus \{0\}$, then $v(ab) = v(a) + v(b) \geqslant v(b) \geqslant 0$, so $v(ab) \notin \Delta$ gives us that $\mathfrak{p}$ is an ideal. Finally, if $a, b \in A \setminus \mathfrak{p}$, then $v(ab) = v(a) + v(b) \in \Delta$ implies $ab \in A \setminus \mathfrak{p}$, whence $\mathfrak{p}$ is prime. It is clear that $\mathfrak{p} \mapsto \Gamma$ under the mapping of $\operatorname{Spec} A$.

The value group of $A_\mathfrak{p}$ is

$$\begin{aligned}
\frac{K^\times}{A_\mathfrak{p}^\times} &= \frac{K^\times}{A_\mathfrak{p} \setminus \mathfrak{p}A_\mathfrak{p}} \\
&= \frac{K^\times/A^\times}{(A_\mathfrak{p} \setminus \mathfrak{p}A_\mathfrak{p})/A^\times} \\
&= \frac{\Gamma}{\pm v(A \setminus \mathfrak{p})} \\
&= \Gamma/\Delta.
\end{aligned}$$

Dually, the value group of $A/\mathfrak{p}$ is

$$\frac{\operatorname{Frac}(A/\mathfrak{p})}{(A/\mathfrak{p})^\times} =$$

□

5.33 Let Γ be a totally ordered abelian group, k any field, and $A := k[\Gamma]$ the group algebra of Γ over k. Show that A is an integral domain.

If $u = \lambda_1\alpha_1 + \cdots + \lambda_n\alpha_n$ is any nonzero element of A, where each $\lambda_i \neq 0$ and $\alpha_1 < \cdots < \alpha_n$, define $v_0(u) := \alpha_1$.[1] Show that $v_0 : A \setminus \{0\} \to \Gamma$ satisfies conditions (a) and (b) of (Ex. 5.31).

Let K be the fraction field of A. Show that v_0 can be uniquely extended to a valuation v of K, and that the value group of v is precisely Γ.

Proof. Take $u, u \in A$ nonzero and write

$$\begin{aligned}
u &= \lambda_1 x_{\alpha_1} + \cdots + \lambda_m x_{\alpha_m}, \\
u' &= \lambda'_1 x_{\alpha'_1} + \cdots + \lambda'_n x_{\alpha'_n}
\end{aligned}$$

[1]This arises from the natural Γ-graded structure of $k[\Gamma]$.

for $\lambda_i, \lambda'_i \neq 0$ and $\alpha_1 < \cdots \alpha_m$, $\alpha'_1 < \cdots \alpha'_n$. Suppose by way of contradiction that

$$\begin{aligned} 0 &= uu' \\ &= \sum_{i=1}^{m}\sum_{j=1}^{n} \lambda_i \lambda'_j x_{\alpha_i+\alpha'_j}. \end{aligned}$$

Then $\alpha_1 + \alpha'_1 < \alpha_i + \alpha'_j$ for all $i, j \neq 1$, so $\lambda_1\lambda'_1 x_{\alpha_1+\alpha'_1} = 0$ implies $\lambda_1 = 0$ or $\lambda'_1 = 0$, a contradiction.

Let $u, u' \in A$ nonzero. By the above, $uu' = \lambda_1\lambda'_1 x_{\alpha_1+\alpha'_1} + \cdots$, so $v_0(uu') = \alpha_1 + \alpha_2$ gives us (a). For (b), we allow $m, n = 0$ and induct on $k = \#\{i \mid \lambda_i + \lambda'_i = 0, \alpha_i = \alpha'_i\}$. If $k = 0$, then clearly $v_0(u + u') = \min\{v_0(u), v_0(u')\}$. If $k = m = n$, then $u + u' = 0$, so the result holds trivially. If only one of $m, n = 0$, then without loss of generality $n = 0$, so $v_0(u+u') = v_0(u) = \min(v_0(u), \infty)$. Otherwise, $v_0(u+u') = v_0(u''+u''')$ for $u'' = u - \lambda_1 x_{\alpha_1}$, $u''' = u' - \lambda_1 x_{\alpha_1}$, and $k' = k - 1$, so

$$\begin{aligned} v_0(u + u') &= v_0(u'' + u''') \\ &\geqslant \min(v_0(u''), v_0(u''')) \\ &\geqslant \min(v_0(u), v_0(u')), \end{aligned}$$

whence the result follows by induction.

Take $K = \operatorname{Frac} A$. Then for any $x, y \in A$ and $y \neq 0$,

$$v(x) = v((xy^{-1})y) = v(xy^{-1}) + v(y)$$

forces $v(xy^{-1}) = v_0(x) - v_0(y)$. Clearly this is a valuation on K, and $v : K^\times \to \Gamma$ is surjective by construction, so Γ is the valuation group of v. $\square$

5.34 Let A be a valuation ring and K its field of fractions. Let $f : A \to B$ be a ring homomorphism such that $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is a closed mapping. Then if $g : B \to K$ is any A-algebra homomorphism (i.e., $g \circ f$ is the embedding $A \hookrightarrow K$), we have $g(B) = A$.

Proof. Set $C := g(B)$. g an A-algebra homomorphism gives us $A \subset C \subset K$, whence $C = A_{\mathfrak{p}}$ for some prime $\mathfrak{p} \leqslant A$ by (Ex. 5.29). Now, consider the maximal ideal $\mathfrak{n} = \ker g \leqslant B$. (Ex. 5.10a) says that f has the going-up property, so $\mathfrak{n}^c \leqslant A$ is maximal. A is local by (5.18i), so $\mathfrak{n}^c = \mathfrak{m}$ is the unique maximal ideal of A.

$g : B \to A_{\mathfrak{p}}$ must map units to units, so $g(\mathfrak{n}) \supset \mathfrak{p}A_{\mathfrak{p}}$. Then $\mathfrak{m} \supset \mathfrak{p}A_{\mathfrak{p}}$

$$\begin{aligned} \mathfrak{p} &= (g \circ f)^{-1}(\mathfrak{p}A_{\mathfrak{p}}) \\ &= f^{-1}(g^{-1}(\mathfrak{p}A_{\mathfrak{p}})) \end{aligned}$$

We have

$$g(\mathfrak{n}) = (g \circ f)(\mathfrak{m}) = \mathfrak{m}$$

$\square$

5.35 From (Ex. 5.1) and (Ex. 5.3), it follows that, if $f : A \to B$ is integral and C is any A-algebra, then the mapping $(f \otimes 1)^* : \operatorname{Spec}(B \otimes_A C) \to \operatorname{Spec} C$ is a closed map.

Conversely, suppose that $f : A \to B$ has this property and that B is an integral domain. Then f is integral.

Show that the result just proved remains valid if B is a ring with only finitely many minimal prime ideals (e.g., if B is Noetherian).

Proof. Suppose first that B is an integral domain, and that $f : A \to B$ has the above property. Take $x \in B$, and consider $C = A[x]$. Then $f^* : \operatorname{Spec} B \to \operatorname{Spec} A[x]$, so $f : A[x] \to B$ has the the going-up property by (Ex. 5.10a).

$\square$